

# The response of reptiles and mammals to fire-driven vegetation succession in semi-arid *Triodia*-mallee woodlands

Jules E. Farquhar<sup>A,B,\*</sup> , Wyn Russell<sup>B</sup>, Jesse B. Farquhar<sup>B</sup>, Simon Cook<sup>B,S</sup> and Ashley R. Olson<sup>B</sup> 

For full list of author affiliations and declarations see end of paper

## \*Correspondence to:

Jules E. Farquhar  
 School of Biological Sciences, Monash University, Clayton, Vic 3800, Australia  
 Email: [Jules.Farquhar@monash.edu](mailto:Jules.Farquhar@monash.edu)

<sup>S</sup>Posthumous author.

## ABSTRACT

**Background.** Predicting faunal responses to fire is complex due to regional differences in fire-vegetation dynamics, necessitating locally calibrated studies. **Aims.** This study examined the effects of fire-mediated vegetation succession on (1) the richness, diversity and composition of small vertebrate communities and (2) individual species abundance in *Triodia*-mallee landscapes. **Methods.** We sampled small vertebrates across 12 sites at Nanya Station, NSW, Australia, with fire histories spanning 1, 18 and >40 years since the last fire (YSLF). **Key results.** 531 vertebrates from 39 species were recorded. There was strong evidence that fire history influenced vegetation structure, which shaped reptile community composition and species abundance. Early successional sites (1 YSLF) supported higher abundances of lizards such as *Ctenotus regius*, *Lerista labialis* and *Lucasium damaeum*. These species peaked 1-year post-fire before declining by 18 YSLF. In contrast, the dasyurid *Ningaui yvonneae* was most abundant 20–40 YSLF, correlating with dense spinifex, and was nearly absent at recently burned sites. **Conclusions.** Fire history drives varied successional responses among species, with some benefiting from early post-fire conditions and others requiring mature spinifex habitat. **Implications.** Region-specific studies are essential for effective conservation and land management. Long-term research is needed to understand successional dynamics and guide fire management strategies.

**Keywords:** Community composition, fire management, habitat structure, lizards, mammals, post-fire recovery dynamics, successional response, *Triodia*-mallee.

## Introduction

Fire is an integral form of landscape-scale disturbance that influences the distribution, richness and abundance of fauna globally (Andersen *et al.* 2005; Bond and Keeley 2005; Bowman *et al.* 2009; York *et al.* 2012; He *et al.* 2019; González *et al.* 2022; Moyo 2022), and is widely used as a management tool for biodiversity conservation (Andersen *et al.* 2005; Edwards *et al.* 2008; Driscoll *et al.* 2010). However, human activities have led to changes in fire regimes, and this presents a risk of extinction to species globally (Bradstock *et al.* 1998; Pausas and Keeley 2009; Griffiths *et al.* 2015; Woinarski *et al.* 2015; Nimmo *et al.* 2022). It is therefore imperative that management decisions are underpinned by robust knowledge of species responses to fire regimes (Driscoll *et al.* 2010).

Many faunal species depend on habitat structural features (Tews *et al.* 2004; Garden *et al.* 2007; Bunce *et al.* 2013; Farquhar *et al.* 2023) and populations often track the dynamic changes in vegetation structure following fire (Friend 1993; Monamy and Fox 2000; Loin and Kennedy 2009; Clarke *et al.* 2021). Fox's (1982) habitat accommodation model (HAM) has widely been used as a conceptual framework for predicting faunal responses to fire-driven secondary succession. The model posits that species will enter the succession when changes in vegetation structure result in a habitat that meets their specific ecological needs (Fox 1982). As community succession continues and conditions become unfavourable for a species, it may reduce in abundance or become eliminated from the succession (Fox 1982). The model suggests that species responses to fire follow predictable patterns of change related to the post-fire regeneration of vegetation, not time-since-disturbance per se (Monamy and Fox 2000; Fox *et al.* 2003; Letnic *et al.* 2004).

**Received:** 3 December 2024

**Accepted:** 26 June 2025

**Published:** 30 July 2025

**Cite this:** Farquhar JE *et al.* (2025) The response of reptiles and mammals to fire-driven vegetation succession in semi-arid *Triodia*-mallee woodlands. *International Journal of Wildland Fire* **34**, WF24208. doi:10.1071/WF24208

© 2025 The Author(s) (or their employer(s)). Published by CSIRO Publishing on behalf of IAWF.

This is an open access article distributed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND)

OPEN ACCESS

In the fire-prone mallee *Eucalyptus* woodlands of southern Australia, where vegetation structure is strongly influenced by time-since-fire (Haslem *et al.* 2011), Caughley (1985) developed a habitat accommodation model specific to the post-fire changes in abundance of desert reptiles. The model recognises three discreet responses to time-since-fire. First, burrowing species are most abundant in early seral stages, second, spinifex grass (*Triodia* spp.) specialists are most abundant beyond 6 years once sufficient *Triodia* has recovered and third, litter dwelling species are most abundant after 25 years once sufficient leaf litter has accumulated (Caughley 1985). Additionally, Driscoll and Henderson (2008) propose that logs are an important resource for reptiles and should be included into Caughley's (1985) model. The temporal availability of these microhabitat resources is expected to have implications for the persistence of fauna in these ecosystems (Haslem *et al.* 2011).

Species responses to fire-mediated habitat changes will depend on their functional traits and how those traits influence interactions with the environment (Sousa 1984). As such, a common approach to fauna-fire studies is to use functional traits related to habitat-use to predict biotic responses to fire (Driscoll and Henderson 2008; Lindenmayer *et al.* 2008; Keith 2012; Nimmo *et al.* 2012). Species that depend on a particular vegetation resource are expected to peak in abundance when that resource becomes widely available after fire (Keith 2012). With this knowledge, it may be possible to extrapolate responses to fire for species with similar ecological traits (Watson *et al.* 2012). These generalisable models could potentially be used to identify seral stages important for species persistence (Watson *et al.* 2012; Clarke *et al.* 2021). However, such generalised response curves have had variable success in predicting species responses to fire (Driscoll and Henderson 2008; Lindenmayer *et al.* 2008; Nimmo *et al.* 2012; Smith *et al.* 2013). Geographical variation in variables unrelated to fire, such as rainfall and grazing history, also drive vegetation changes (Letnic *et al.* 2004), causing fauna-fire relationships to vary regionally (Nimmo *et al.* 2012; Smith *et al.* 2013; Nimmo *et al.* 2014). Thus, to achieve region specific conservation and land management goals, we need to know whether fire management actions are best guided by generalised species response models (Watson *et al.* 2012) or by knowledge of local fauna-fire relationships that may be idiosyncratic to the site being managed.

In this study, we explored the response of small terrestrial vertebrates (reptiles and mammals) to fire-driven vegetation succession in semi-arid Australia, where lizards comprise a remarkably abundant and diverse taxonomic group among faunal communities (Morton and James 1988). Because reptiles respond to fire over decadal and centennial time scales (Nimmo *et al.* 2012; Valentine *et al.* 2012), we examined fire histories up to 40 years using a space-for-time substitution approach where community data are collected from areas known to have burnt at different times (Pianka and Goodyear 2012; Smith *et al.* 2013). First, we investigated the effect of

fire history on the richness, diversity and composition of vertebrate communities and the abundance of common species. We then investigated whether changes in the abundance of a species could be explained by changes in habitat structure associated with fire-driven vegetation succession.

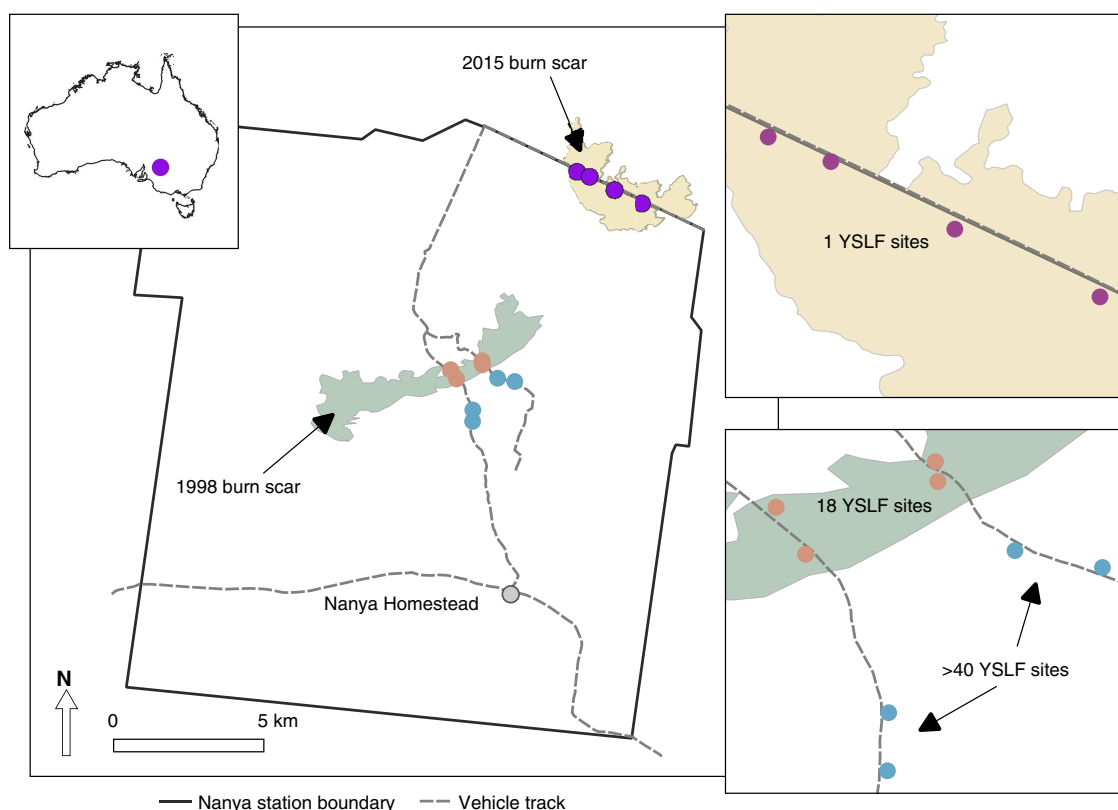
## Materials and methods

### Study area

The study was conducted at Nanya Station, a 40,000 ha ex-pastoral property in the Scotia mallee region of far western NSW (33°7'48.70"S, 141°17'42.04"E) managed for conservation and teaching by Federation University Australia. The climate is semi-arid with a mean annual rainfall of 250 mm but can be highly variable between years. Summer temperatures are high (February mean daily max = 32°C, min = 16°C) while winters are mild (July mean daily max = 15°C, min = 5°C) (Westbrooke 2010). The area comprises relatively intact vegetation communities with no recent livestock grazing in the last 40 years (Westbrooke 2012). The most widespread vegetation community is *Eucalyptus* open-shrubland with *Triodia scariosa* understorey (*Triodia*-mallee) that occurs on low, undulating, sand dunes (Westbrooke *et al.* 1998). The fire history of Nanya has resulted in a mosaic of age classes spanning a century-long timeframe. Extensive wildfire occurred throughout much of the region in 1917 (Withers 1989) and large areas were burnt during 1976 (Rodda 1978), followed by a 5000 ha fire in December 1997 (Westbrooke 2010). A more recent 390 ha wildfire occurred on the northern boundary of Nanya Station in November 2015.

### Sampling

We used a space-for-time substitution to examine changes in reptile community structure along a 40-year post-fire chronosequence. We sampled from three discreet fire histories; 1-year since the last fire (burned in 2015), 18 years since the last fire (burned in 1998) and greater than 40 years since the last fire (burned in 1976 or earlier) (Fig. 1). Four survey sites were established within each treatment, equating to a total of 12 survey sites. All sites were placed approximately mid-dune, on sandy soils in *Triodia*-mallee vegetation between 50 and 89 m elevation and spaced at least 250 m apart to prevent pseudoreplication. Sampling was conducted over four 12-day fieldtrips across two Austral spring-summer periods (December 2016, February 2017, September 2017 and January 2018) using pit-fall and funnel trapping with a total sampling effort of 4144 trap nights. Importantly, every site was sampled the same number of times and on the same days, thereby removing the effects of survey time period, weather conditions and sampling effort on inter-site comparisons. This enabled us to use the sum total of captures at each site as the sampling unit. A 25 m trapping line was constructed at each site, consisting of five



**Fig. 1.** Map of Nanya station showing the location of landscape fires that occurred in 1998 and 2015. Also shown is the location of each sampling site (1 year, 18 years and greater than 40 years since the last fire (YSLF)).

20 L buckets spaced at 4 m intervals and connected by a 30 cm high flywire fence. Two funnel traps (Professional Trapping Supplies, Australia. Trap dimensions: 18 cm × 18 cm × 79 cm) were placed in the centre of the trapping line, one either side of the drift fence. Traps were checked daily. After taking basic morphological measurements, animals were marked on the ventral surface with non-toxic permanent marker to distinguish recaptures. Animals were released at the site of capture within 24 h, at least 20 m from the trapping line to minimise short-term recaptures.

## Vegetation assessment

To quantify structural differences in habitat among fire ages, ground cover and understorey structure were assessed at each site using a structure 'touch-pole' over a 100-point grid (1 m intervals between points). Ground cover attributes contacting the base of the pole were recorded as either *Triodia*, litter/debris or bare ground. Vertical vegetation density was assessed by recording the number of understorey structural contacts with the pole up to 2 m high. Vertical vegetation structure contacts were grouped into two categories: *Triodia* and non-*Triodia* vegetation density (shrubs, mallee, herbs and tussock grasses). Canopy cover was quantified for each site by one observer placing a circular cardboard tube over one eye and looking directly

upward to visually estimate the percentage of canopy foliage encroaching on the circular view. All vegetation above the 2 m touch-pole was considered to be canopy foliage.

A principal component analysis (PCA) was performed on vegetation variables to avoid collinearity and simplify analytical models. All variables were centred and scaled prior to conducting the PCA and structural density variables (*Triodia*, other vegetation <1 m and other vegetation 1–2 m) were square-root transformed. The resulting principal component (PC) dimensions used for subsequent analysis were selected by confirming that the variation captured in the dimension was significantly different from random variation in the data using the *PCtest* package.

## Data analysis

We investigated the relationship between the number of years since the last fire (YSLF) and species richness, Shannon's diversity index ( $H'$ ), community composition (*beta* diversity), vegetation structure (PCA dimensions) and the abundance of the most common species (species that were observed >20 times in total). To do so, we fitted generalised linear models (GLMs) in a Bayesian framework using the *brm* function from the *brms* package in R (Bürkner 2017; R Development Core Team 2024) with YSLF as a categorical predictor variable with a level for each YSLF category (1, 18

and >40). We also investigated the effect of vegetation structure on the abundance of the most common species using Poisson GLMs using PCA dimensions of structure as predictor variables. All models were fitted using default priors and three Markov chain Monte Carlo (MCMC) simulations with 4000 iterations, after the first 1000 were discarded as burn-in, for a total of 12,000 iterations. Poisson GLMs (PGLM) or negative binomial GLMs (NBGLM) were used to determine support for relationships between response variables that were count data (richness and species abundance) and YSLF, depending on whether the data were overdispersed. For continuous response variables ( $H'$  and PCA dimensions), we fitted Gaussian GLMs. For each model, we plotted parameter estimates and provided highest posterior density (HPD) intervals as an indication of uncertainty. Model diagnostics were performed by checking that Rhat values were less than 1.01, MCMC effective sample size was greater than 2000, and by performing graphical posterior predictive checks using the *pp\_check* function. Finally, pairwise comparisons between YSLF categories were performed using the *pairs* function from the *emmeans* package which calculates the difference in estimated means and associated HPD intervals (Lenth 2025).

Community composition was compared among burn treatments using a PerMANOVA via the *adonis* function in the *vegan* package in R (Oksanen 2024). PerMANOVA is a distance-based nonparametric multivariate analysis that provides a pseudo  $F$ -statistic value and derives a  $P$ -value from permutation tests. PerMANOVA was performed using square-root transformed abundance data and a Bray–Curtis dissimilarity matrix. We used 10,000 random permutations to test for significant variation in community composition among different YSLF categories. A groupwise comparison among YSLF treatments was initially performed, followed by pair-wise comparisons to detect differences between each pair of treatments (Anderson 2001). The Bonferroni correction was applied to  $P$ -values calculated from pairwise comparisons to avoid false discoveries. Dissimilarities were visualised to illustrate reptile assemblage trends using non-metric multidimensional scaling (nMDS) among treatments using the *metaMDS* function in the *vegan* package (Oksanen 2024).

## Ethics

This research was conducted with approval from the Federation University Animal Ethics Committee (Project Number: 16-007), and in accordance with a NSW scientific research permit (Permit: SL100193).

## Results

Pit-fall trapping at the 12 sites resulted in a total of 531 captures comprising 34 reptile species, four mammals and one amphibian (Appendix 1). The most captured species were the dasyurid *Ningauy yvonnae* ( $n = 129$ ), the agamid

lizard *Ctenophorus spinodomus* ( $n = 113$ ), the skinks *Ctenotus atlas* ( $n = 34$ ), *Ctenotus regius* ( $n = 31$ ) and *Lerista labialis* ( $n = 29$ ), and the gecko *Lucasium damaeum* ( $n = 20$ ). Fourteen species were captured less than three times; of those, 12 were reptile species (six snakes, three geckos, two skinks and one pygopod), one was a mammal, *Cercartetus concinnus*, and one was the amphibian, *N. sudelae*. Four reptile species and three mammal species are considered threatened in NSW (Appendix 1).

## Response of community richness, diversity and composition to YSLF

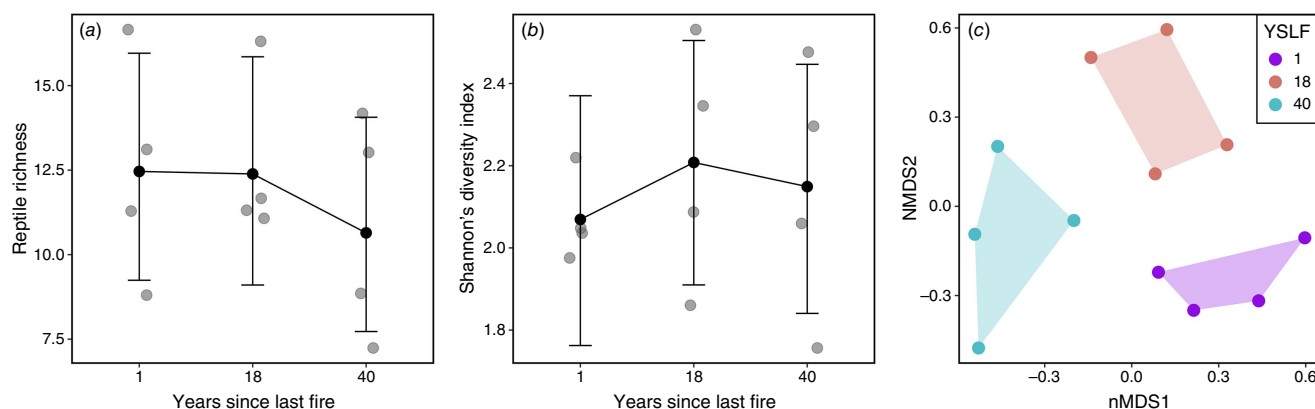
The overall mammal assemblage was too depauperate to investigate patterns of richness, diversity and composition (total richness = 4). Thus, we restricted our investigation of the ecological community to the reptile assemblage (Fig. 2). Pairwise comparisons revealed no support for differences between YSLF categories in species richness or Shannon's diversity index (95% HPD included zero in all cases; Table 1). However, YSLF significantly influenced the composition of the reptile community (three-way PerMANOVA:  $F = 3.512$ ,  $P < 0.001$ ) (Fig. 2). Pairwise PerMANOVA comparisons indicated that the composition of species was significantly different between sites that were 1-year post-fire and 18-years post-fire ( $F = 3.561$ ,  $P = 0.029$ ), 1-year and > 40-years post-fire ( $F = 4.619$ ,  $P = 0.029$ ), and 18-years and >40 years post-fire ( $F = 2.46$ ,  $P = 0.031$ ).

## Influence of YSLF on abundance

The six most commonly trapped vertebrate species varied in their response to fire, from 'irruptive' (high early succession abundance) to 'plateau' (abundance increases to an asymptote in the intermediate-late succession) and 'null' (Clarke et al. 2021; Fig. 3). The abundance of the agamid species *C. spinodomus* (the most frequently trapped reptile) showed a decline in abundance with YSLF, with strong evidence the species was more abundant 1-year post-fire than at >40-years post-fire (Table 1). The skinks *C. regius* and *L. labialis*, and the gecko *L. damaeum*, all showed a similar preference for recently burned sites; these species showed strong evidence that they were most abundant at the 1-year post-fire sites then displayed a substantial decline (i.e. irruptive response), more attenuated than that of *C. spinodomus*, between 1-year and 18-years post-fire (Table 1). There was no evidence that the abundance of *C. atlas* differed according to the fire history of a site (Table 1).

The most common mammal species, *N. yvonneae* showed for a positive relationship with YSLF (Fig. 3). This species was rarely observed at sites that had burned 1-year prior to our survey (total observations = 2) and there was strong evidence that it was substantially more abundant at sites that were 20–40 years post-fire (Table 1). In total, *N. yvonneae* was observed 63 times at sites that had burned 18 years





**Fig. 2.** The effect of fire history (years since last fire (YSLF)) on (a) the richness, (b) diversity and (c) composition of reptile communities. For richness and diversity, points indicate predicted values of Poisson and Gaussian models, respectively, and error bars indicate 95% highest posterior density (HPD) intervals. Changes in community composition are demonstrated by a non-metric multidimensional scaling (nMDS) biplot showing dissimilarity of reptile community composition at sample sites 1, 18 and >40 years after being burned. Bray–Curtis dissimilarity was used to quantify the differences in community structure between each sample site.

prior to our study and 64 times at sites that had burned 40 years or more before our study. This indicates that the abundance of *N. yvonneae* increases to a plateau over a 40-year chronosequence.

### Effect of vegetation on abundance

The first dimension (PC1) of the principal component analysis explained 57.3% of the variation in the habitat structure data and was the only dimension that was significantly different from random variation ( $P < 0.05$ ). PC1 revealed a clear separation among fire age classes in vegetation structure (Fig. 4) as indicated by strong evidence for differences between the 1 YSLF treatment and both the 18 YSLF treatment and >40 YSLF treatment (Table 1). However, there was no evidence for a difference in PC1 between the 18 YSLF and the >40 YSLF treatments (Table 1). The differences in PC1 values between seral stages indicates that sites 1-year post-fire had greater coverage of bare ground and greater density of low (0–1 m high) non-*Triodia* vegetation (e.g. small herbs regenerating after the fire). By comparison, sites that were 18 or >40 years post-fire showed greater canopy cover, more woody debris on the soil surface and greater *Triodia* and mid-storey vegetation density.

Small vertebrates showed variable responses to the vegetational changes driven by fire (Fig. 5). Strong evidence for a positive correlation with PC1 was observed for four species of lizard; *C. spinodomus* (PGLM:  $\beta = 0.23$ , 95% HPD: [0.14, 0.33]), *C. regius* (PGLM:  $\beta = 0.80$ , 95% HPD: [0.55, 1.08]), *Lerista labialis* (PGLM:  $\beta = 0.49$ , 95% HPD: [0.29, 0.70]), and *Lucasium damaeum* (PGLM:  $\beta = 0.43$ , 95% HPD: [0.19, 0.68]). Positive correlations with PC1 indicate that the aforementioned four species preferred sites with relatively more bare ground and understorey vegetation that did not include *Triodia*. In comparison, there was no evidence for a relationship between PC1 and the abundance of *C. atlas*

(PGLM:  $\beta = -0.13$ , 95% HPD: [-0.35, 0.07]). As the abundance of *C. atlas* was not correlated with YSLF either, it appears that the abundance of this species is not influenced by fire-driven vegetation succession at the fire intervals and locations we sampled. The mammal, *N. yvonneae*, showed strong evidence for a negative correlation with PC1 (PGLM:  $\beta = -0.53$ , 95% HPD: [-0.70, -0.38]), indicating a preference for sites with greater canopy cover and woody debris, as well as a higher density of *Triodia* and mid-storey vegetation.

### Discussion

Regional variability in fire regimes, biota, climate and landscape factors can make generalisations in fire ecology problematic (Keith 2012; Nimmo *et al.* 2013; Connell *et al.* 2022). Instead, fire management practices might best be guided by region-specific knowledge of vegetation communities to optimise habitat for fauna; however, such information is not available for most regions, and obtaining detailed region-specific data are impractical for board-scale fire management (Driscoll *et al.* 2010). Nonetheless, such arduous approaches may be necessary to meet the precision required for threatened species management, because fire regimes guided by inaccurate generalisations could have dire consequences for biodiversity. Although our study was small in scale, it can benefit local management and serves as a reference point for comparison with other regions and future assemblages that may experience different climates and fire-vegetation dynamics.

### Vegetation-fire relations at Nanya Station

Clear successional changes in habitat structure were observed after fire. The most conspicuous change is the recovery of *Triodia* hummocks and the consequent decline in bare ground

**Table 1.** Pairwise contrasts between years since last fire categories for each response variable.

| Response             | Contrast | $\Delta$ estimate | 95% HPD intervals |
|----------------------|----------|-------------------|-------------------|
| Richness             | 1–18     | 0.00              | –0.37, 0.41       |
|                      | 1–40     | 0.16              | –0.25, 0.56       |
|                      | 18–40    | 0.15              | –0.27, 0.55       |
| Diversity            | 1–18     | –0.14             | –0.56, 0.28       |
|                      | 1–40     | –0.08             | –0.51, 0.34       |
|                      | 18–40    | 0.06              | –0.36, 0.49       |
| <i>C. spinodomus</i> | 1–18     | 0.70              | –0.26, 1.65       |
|                      | 1–40     | 1.16              | 0.16, 2.09*       |
|                      | 18–40    | 0.46              | –0.54, 1.47       |
| <i>C. atlas</i>      | 1–18     | –0.65             | –1.56, 0.19       |
|                      | 1–40     | –0.33             | –1.26, 0.61       |
|                      | 18–40    | 0.32              | –0.49, 1.11       |
| <i>C. regius</i>     | 1–18     | 3.66              | 1.73, 6.30*       |
|                      | 1–40     | 11.94             | 2.51, 31.20*      |
|                      | 18–40    | 8.22              | –2.19, 28.10      |
| <i>L. labialis</i>   | 1–18     | 1.39              | 0.49, 2.28*       |
|                      | 1–40     | 9.34              | 1.81, 26.99*      |
|                      | 18–40    | 7.94              | 0.35, 25.76*      |
| <i>L. damaeum</i>    | 1–18     | 1.38              | 0.31, 2.61*       |
|                      | 1–40     | 3.00              | 0.89, 5.51*       |
|                      | 18–40    | 1.62              | –0.59, 4.51       |
| <i>N. yvonneae</i>   | 1–18     | –3.55             | –5.29, –2.26*     |
|                      | 1–40     | –3.57             | –5.29, –2.25*     |
|                      | 18–40    | –0.02             | –0.37, 0.34       |
| Vegetation (PC1)     | 1–18     | 3.44              | 2.15, 4.73*       |
|                      | 1–40     | 3.54              | 2.25, 4.82*       |
|                      | 18–40    | 0.11              | –1.18, 1.40       |

Poisson generalised linear models were used for all analyses, except for *C. spinodomus*, where a negative binomial generalised linear model was used. Estimates and highest posterior density (HPD) intervals are reported on a log scale meaning strong evidence for a difference between groups is indicated by HPD intervals excluding zero and is denoted by an asterisk. PC1 refers to principal component 1.

cover over time. Recently burnt (1 YSLF) areas have extensive bare ground coverage, and the only available form of above-ground vegetation cover is small regenerating herbaceous plants. At 18 YSLF, *Triodia scariosa* tussocks become large and dense, while mallee *Eucalyptus* spp. had regrown to >2 m tall. At 40 years, the cover of *Triodia* climaxes, forming large and connected hummocks (Rice and Westoby 1999). Similar patterns of vegetation recovery have been noted in other studies of *Triodia*-mallee communities (Caughley 1985; Haslem et al. 2011; Clarke et al. 2021).

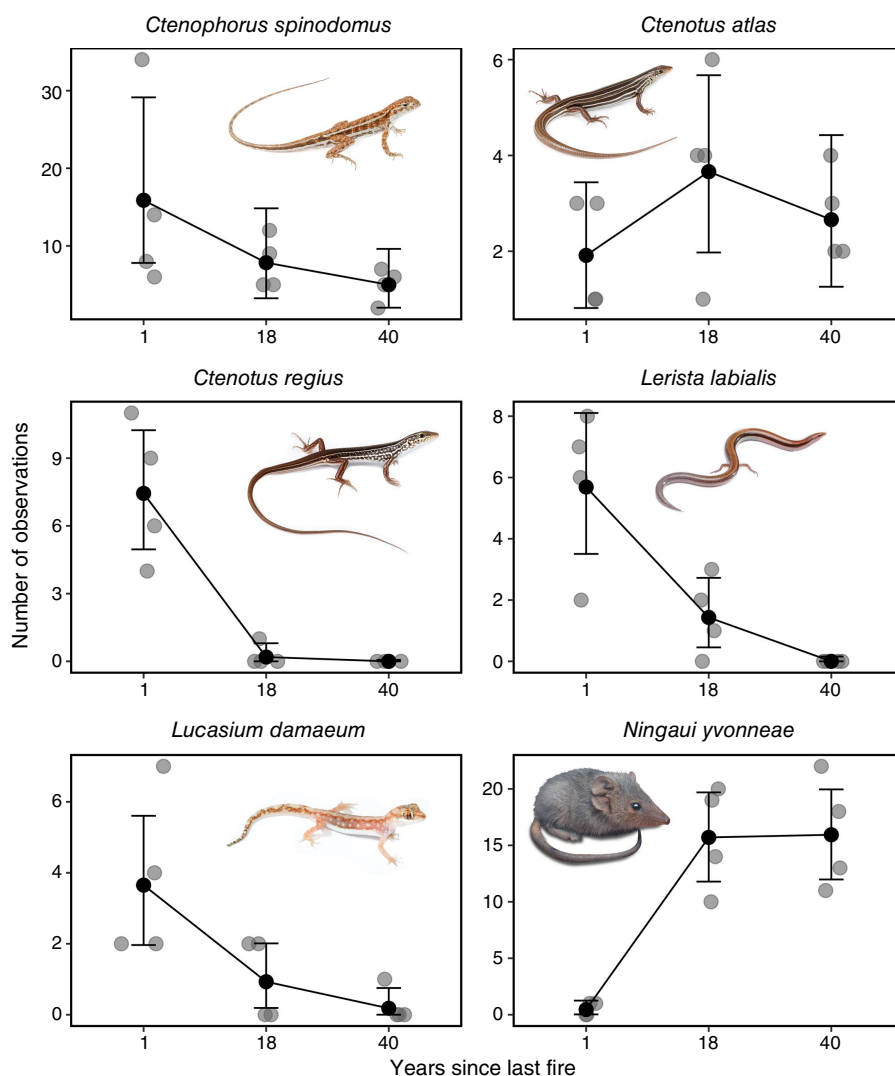
## Fauna-fire relationships at Nanya station

Despite strong evidence for successional changes in habitat structure, reptile richness and diversity did not vary according to fire history at our sites. This observation is consistent with studies in other ecosystems on reptile richness in south-western Australia, including in *Banksia* woodlands (Davis and Craig 2024) and temperate jarrah forests (Doherty et al. 2024a), and in a range of vegetation types in eastern Australia (Lindenmayer et al. 2008; Hu et al. 2013; Archer et al. 2024). Conversely, in a sclerophyll forest of southeastern Australia, Dixon et al. (2018) showed that reptile richness and abundance are significantly higher in sites >96 years post-fire than younger fire ages (0.5–12 years). Inconsistency between reptile richness and fire history was noted in a recent meta-analysis by González et al. (2022). Further, a global meta-analysis of organismal responses to fire (Moyo 2022) showed that fire had no effect on abundance and diversity of reptiles but had a small positive effect on richness of reptiles.

Reptiles can display strong turnover in community composition following fire, which likely masks variation in reptile species richness because late successional specialists are quickly replaced by early successional species (Doherty et al. 2024b). We found that reptile community composition differed in response to fire, with each of the three fire age classes supporting a substantially different assemblage of species. This supports an extensive base of literature demonstrating that fire-driven vegetation succession is a key driver of variation in reptile community assembly (Caughley 1985; Greenberg et al. 1994; Masters 1996; Letnic et al. 2004; Valentine et al. 2012; Clarke et al. 2021; González et al. 2022; Partridge et al. 2023).

Although PC1 suggests that broad vegetation structure is similar in 18 and >40 YSLF sites, reptile communities remain compositionally distinct. We attribute this to successional changes in ecological dynamics not measured in this study, but which may continue to diverge beyond 18 years post-fire. Older sites (e.g. >40 YSLF) may continue to shift in finer-scale microhabitat features and invertebrate prey communities that are not captured by PC1, but underlie the nMDS-identified turnover between mid- and late-successional sites. Future studies should integrate metrics of prey availability, debris, litter depth, soil properties and moisture to fully explain the associations between fauna and fire history in these ecosystems.

As our models of individual species capture rates (Fig. 3) are limited to the six most frequently captured species, they cannot account for the rarer taxa that contribute substantially to the community-level differences shown in the nMDS (Fig. 2c). While our individual-species models clearly distinguish 1 YSLF from both 18 YSLF and >40 YSLF, they do not reveal any contrasts between the two older age classes. The nMDS, by including all species – even those too infrequent to model – demonstrates that 18 YSLF and >40 YSLF plots



**Fig. 3.** The effect of fire on the abundance of the six most common species at the 12 sample sites. Black points represent predicted values and error bars indicate 95% highest posterior density intervals based on Poisson generalised linear models. Grey points (observed values) are slightly transparent to show overlapping data more easily.

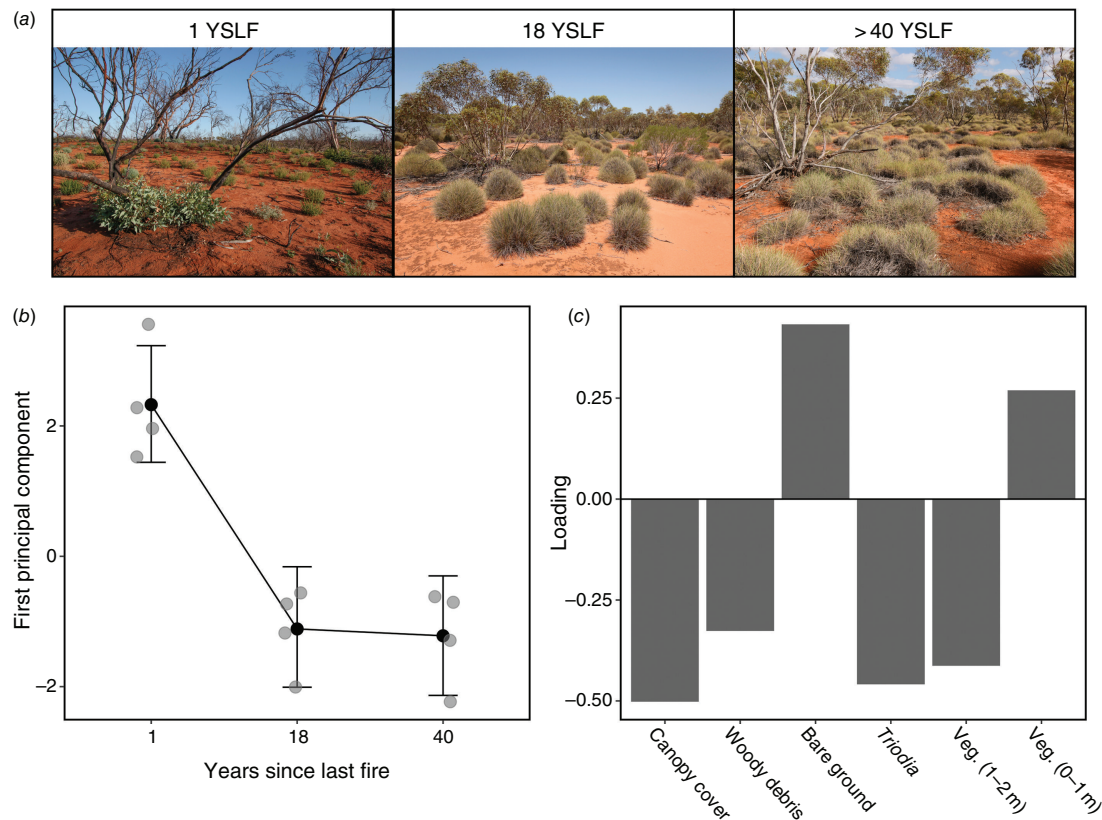
are themselves distinct. We therefore infer that low-abundance species underlie the divergence between these older treatments. Targeted surveys and modelling of these rare taxa will be essential to fully understand how fire-history shapes the entire reptile assemblage.

### Individual species responses

A range of animal functional traits influence a species' fire sensitivity and therefore response to fire (Lazzari *et al.* 2022; Batista *et al.* 2023). We explored responses by comparing relative abundance between sites, under the assumption that higher abundance in a given burn age is indicative of a species' preference for that age class. Here, we discuss the fire-response of the six most commonly trapped species and compare them to studies from other regions to understand the degree of inter-regional variation.

Three common, terrestrial burrowing species (*L. labialis*, *C. regius* and *L. damaeum*) displayed significant positive

relationships with PC1, peaking in abundance at recently burned (1 YSLF) sites, characterised by high bare ground cover and lack of *Triodia*. This is consistent with the findings of similar studies (Fyfe 1980; Caughley 1985; Sass and Wilson 2006; Driscoll and Henderson 2008) and aligns with the 'irruptive' fire response type – species that burrow or favour open spaces for foraging display greatest frequency of occurrence in early post-fire habitat (Caughley 1985; Friend 1993; Clarke *et al.* 2021). *Lerista labialis* is a sand-swimming specialist, spending much of its time in the loose upper surface of the sand (Greenville and Dickman 2009), and hence may be relatively unaffected by the loss of biomass following fire. Greenville and Dickman (2009) experimentally show that site selection in *L. labialis* is largely driven by a preference for loose sandy soil, rather than vegetation, food availability or thermal regimes of a site. Such a preference may interact with fire history, given that fire destroys the soil crust and enables loose sand to redistribute (Greene *et al.* 1990), thereby creating more extensive burrowing opportunities for



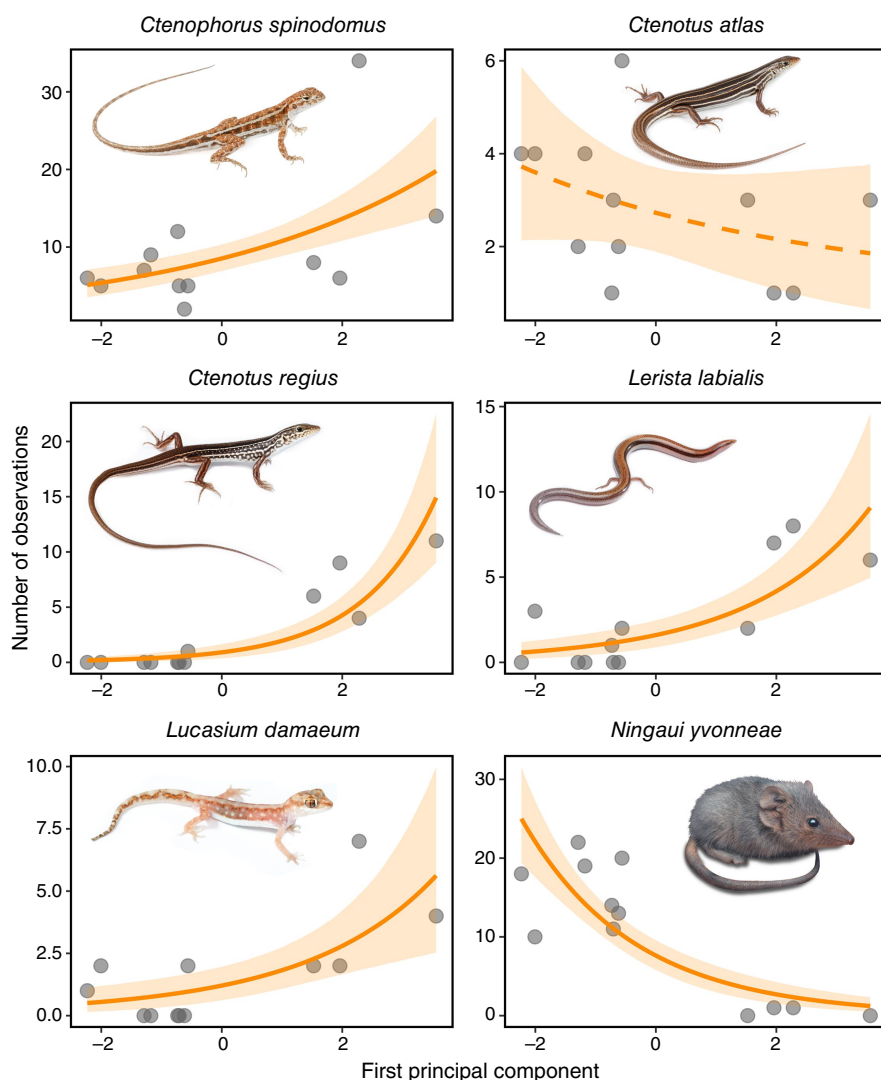
**Fig. 4.** Changes in habitat structure following fire. Photos (a) show examples of the vegetation 1-year, 18 years and >40 years since last fire (YSLF). (b) The relationship between first principal component and fire history (YSLF), where black points represent predicted values and the error bars indicate 95% highest posterior density intervals based on a Poisson generated linear model. Grey points (observed values) are slightly transparent to show overlapping data more easily. (c) Principal component analysis loadings for each habitat variable.

fossorial squamates like *L. labialis* (Greenville and Dickman 2009). Our evidence for higher abundance of *L. labialis* at recently burnt sites indicates that fire benefits this common species by removing soil-stabilising crusts and vegetation. *Ctenotus regius* is a generalist species found in a range of habitat types and regions that lack *Triodia*. *Ctenotus regius* shelters in burrows it constructs into soil, and is likely minimally affected by the loss of complex vegetation structures following fire. At Nanya Station, *Ctenotus regius* has been recorded eating the inflorescence of small herbaceous plants (Olson *et al.* 2018) which are markedly abundant at 1 YSLF sites. It is possible that occasional herbivory in *Ctenotus regius* is a trait that may enhance its capacity to persist in recently burnt sites compared to other more strictly insectivorous and *Triodia* dependent species. *Lucasium damaeum* forges widely in open spaces and shelters in disused burrows made by other animals (Henle 1990) – again, a strategy that likely enables this species to persist in the post-fire environment where vegetation is lacking. In mallee eucalypt woodlands, gecko body condition and growth rates are correlated with increased early successional invertebrate abundance (Smith 2018) suggesting that trophic interactions may influence

successional changes in abundance of insectivorous geckos such as *L. damaeum*.

The response of *C. spinodomus* (a non-burrowing spinifex specialist) was consistent with the ‘decline’ fire response type (Clarke *et al.* 2021), showing a decline in occurrence with increasing time since fire, but in a more gradual manner than the irruptive response type. This suggests that, over successional time, the habitat becomes increasingly less suitable for this species in this ecosystem. Others have shown that *C. spinodomus* (formerly *C. fordii* in older studies in NSW) has highest capture rates in early successional vegetation (Cogger 1969, 1974; Caughley 1985; Avitabile 2014). Conversely, Nimmo *et al.* (2012) examined reptile responses to fire over a 100 year chronosequence, and found that *C. spinodomus* (=‘*C. fordii* northern subregion’ in that study) had a preference for sites 50–100 years post-fire. Spatial differences in rainfall and grazing can influence fire-vegetation dynamics, and this has been posited as an explanation for the varied responses of reptiles to fire (Nimmo *et al.* 2012, 2014). *Ctenophorus spinodomus* is widely regarded as a spinifex-obligate species (Cogger 1969, 1974; Caughley 1985; Schlesinger *et al.* 1997; Sadler *et al.* 2019) but observations





**Fig. 5.** The response of five lizards and one mammal (*N. yvonneae*) to habitat structure at the 12 sample sites. Structure was quantified by six habitat variables that were reduced to a single dimension (the first principal component (PC1)) via a principal component analysis (see Fig. 4 for variables and principal component loadings). For all plots, solid lines indicate strong evidence for a relationship between abundance and PC1 (slope highest posterior density (HPD) intervals do not include zero on a log scale) and ribbons represent HPD intervals. Lizard photos by Jules Farquhar; mammal photo by Kris Bell.

of the species occurring in highest abundance at recently burned sites lacking *Triodia* requires explanation. It is possible that, although closely associated with spinifex ecosystems, and frequently using *Triodia* hummocks, the species is not so strictly dependent on them. Indeed, Bell *et al.* (2021) experimentally showed that *C. spinodomus* displays considerable intra-specific variation in the usage of *Triodia* over other vegetation, from total avoidance in some individuals (1.3% use) to exclusive use (100%) in others. Trophic dynamics may also play a role in this species' high abundance at recently burned sites. For instance, *C. spinodomus* feeds almost exclusively on ants (Sadler *et al.* 2019), which can display a marked increase in their surface activity soon after fire (Andersen and Yen 1985). Given these observations, we suggest that *C. spinodomus* is an overall abundant species in part because of its ecological versatility, and this may contribute to explaining why past studies have found varying fire responses in this species. Further research on the post-fire resource use of *C. spinodomus* is needed before one confidently claims that it prefers a certain seral stage.

Although *C. atlas* counts peaked in the 18 YSLF plots (15 individuals versus 8 in 1 YSLF and 11 in >40 YSLF), neither years-since-fire nor the *Triodia*-cover axis (PC1) predicted these differences statistically. This pattern is surprising given the species reliance on *Triodia* hummocks for foraging and shelter (Pianka 1969; Craig *et al.* 2006; Bell *et al.* 2021) and the strong positive relationships reported elsewhere (Caughley 1985; Driscoll and Henderson 2008; Driscoll *et al.* 2012; Smith *et al.* 2013; Verdon *et al.* 2020). Early post-fire environments tend to lack the mature hummocks this skink prefers, yet we still detected them in such environments – perhaps reflecting transient persistence or the species' overall abundance and resilience to fire. Targeted sampling across other successional classes (e.g. 5–10 YSLF) will be essential to resolve more precisely how *C. atlas* tracks hummock regeneration over time at Nanya Station.

*Ningaui yvonneae* showed a 'plateau' response to fire history that is typical of a spinifex specialist (Clarke *et al.* 2021), being virtually absent from 1 YSLF sites, and becoming very abundant between 18 and 40 YSLF sites. This finding is

comparable with previous studies demonstrating the species' preference for sites older than ~20 YSLF (Kelly *et al.* 2010, 2011; Clarke *et al.* 2021), suggesting that accurate inter-regional predictions of this species' occurrence and abundance can be derived from fire history and vegetation data. For instance, Verdon *et al.* (2020) show that *N. yvonneae* returns to the succession when a site's *Triodia* cover is extensive and >20 cm tall (a feature of older sites). Preserving patches of long-unburnt spinifex habitat would benefit the conservation of this species.

## Management implications

Our findings suggest that some common burrowing reptiles are able to tolerate a range of seral stages, and the need to provide a particular vegetation age class may not be as urgent for such species. Others, such as *Ningaui yvonneae* (Vulnerable in NSW), clearly require old growth habitat with mature *Triodia*. Given the variability in fauna responses to fire, both inter-regionally and interspecifically, we urge that a careful management of fire is required at Nanya Station.

Specific fire management recommendations for Nanya are difficult to derive from the present study alone, for several reasons. First, we focused on small terrestrial vertebrates (primarily reptiles), hence separate and tailored fire management actions must be considered for birds, mammals and invertebrates. Second, although time-since-fire has a major effect on fauna in many studies (reviewed in González *et al.* 2022), these fauna-fire responses can be difficult to disentangle from the interacting effects of animal dispersal capabilities (Nimmo *et al.* 2019), habitat loss and fragmentation (Driscoll *et al.* 2021), the timing of fire in relation to prevailing climatic conditions (Connell *et al.* 2022), fire severity, burn area size and vegetation type (Miritis *et al.* 2024). Given this, we provisionally make the general recommendation that a fine-grained mosaic of fire age classes should be a management priority, as each fire age class examined in this study had significantly different community composition (Fig. 2). Recent fire simulations by French *et al.* (2024) in mallee eucalypt woodlands support this management action, showing that small fires form fine-grained mosaics with a stable habitat mixture and with habitat diversity occurring at fine scales.

Aside from maximising *beta* diversity in a landscape, another critical conservation aim is to tailor fire management around the most fire-sensitive and threatened species. Such species may be especially vulnerable to fire due to their low population size and restricted ranges (Connell *et al.* 2019). Many threatened species depend on critical habitat attributes that are often most available in long-unburnt habitats, such as coarse woody debris or large grass hummocks; hence, maintaining long-unburnt habitat is critical for conserving most threatened species (von Takach *et al.* 2022). This is likely the case for small vertebrates at Nanya Station also – in our study, four reptile species and three mammal

species are considered threatened in NSW, most of which are spinifex-obligate species (i.e. *Strophurus elderi*, *Delma australis*, *Cyclodomorphus melanops elongatus*, *Ningaui yvonneae*). Unfortunately, some of these species can be difficult to detect by virtue of their rarity and cryptic habitat-use patterns (e.g. *S. elderi* and *C. m. elongatus* rarely leave spinifex), and hence their fire responses are difficult to quantify without a more targeted sampling strategy. Most studies, including ours, are dominated by a small number of common species, allowing only a small portion of the assemblage to be modelled (e.g. Schlesinger *et al.* 1997), leaving many species' fire responses unknown. We encourage more detailed and ongoing investigations into fauna-fire relations at Nanya Station in order to further refine management decisions. Until then, priority should be placed on maximising pyrodiversity (French *et al.* 2024), but with emphasis on maintaining the availability of key habitat features that are found only in long-unburnt habitats (Dixon *et al.* 2018; von Takach *et al.* 2022).

## References

- Anderson MJ (2001) A new method for non-parametric multivariate analysis of variance. *Austral Ecology* 26, 32–46. doi:10.1111/j.1442-9993.2001.01070.pp.x
- Andersen AN, Yen AL (1985) Immediate effects of fire on ants in the semi-arid mallee region of north-western Victoria. *Australian Journal of Ecology* 10(1), 25–30. doi:10.1111/j.1442-9993.1985.tb00860.x
- Andersen AN, Cook GD, Corbett LK, Douglas MM, Eager RW, Russell-Smith J, Setterfield SA, Williams RJ, Woinarski JC (2005) Fire frequency and biodiversity conservation in Australian tropical savannas: implications from the Kapalga fire experiment. *Austral Ecology* 30, 155–167. doi:10.1111/j.1442-9993.2005.01441.x
- Archer ML, Letnic M, Murray BR, Webb JK (2024) After the megafires: effects of fire severity on reptile species richness and occupancy in South-Eastern Australia. *Fire* 7, 349. doi:10.3390/fire7100349
- ASH (2023) Australian Society of Herpetologists Official List of Australian Species. Available at <https://www.australiansocietyofherpetologists.org/ash-official-list-of-australian-species> [verified 1 December 2024].
- Avitabile S (2014) Mallee Hawkeye Project: Final Report 2011–2014. (Department of Environment and Primary Industries: Melbourne, Vic, Australia)
- Batista EK, Figueira JE, Solar RR, de Azevedo CS, Beirão MV, Berlinck CN, Brandão RA, de Castro FS, Costa HC, Costa LM, Feitosa RM, *et al.* (2023) In case of fire, escape or die: a trait-based approach for identifying animal species threatened by fire. *Fire* 6(6), 242. doi:10.3390/fire6060242
- Bell KJ, Doherty TS, Driscoll DA (2021) Predators, prey or temperature? Mechanisms driving niche use of a foundation plant species by specialist lizards. *Proceedings of the Royal Society B. Biological Sciences* 288, 20202633. doi:10.1098/rspb.2020.2633
- Bond WJ, Keeley JE (2005) Fire as a global “herbivore”: the ecology and evolution of flammable ecosystems. *Trends in Ecology & Evolution* 20, 387–394. doi:10.1016/j.tree.2005.04.025
- Bowman DM, Balch JK, Artaxo P, Bond WJ, Carlson JM, Cochrane MA, D'Antonio CM, DeFries RS, Doyle JC, Harrison SP, Johnston FH (2009) Fire in the earth system. *Science* 324, 481–484. doi:10.1126/science.1163886
- Bradstock RA, Bedward M, Kenny BJ, Scott J (1998) Spatially-explicit simulation of the effect of prescribed burning on fire regimes and plant extinctions in shrublands typical of south-eastern Australia. *Biological Conservation* 86, 83–95. doi:10.1016/S0006-3207(97)00170-5
- Bunce RGH, Bogers MM, Evans D, Halada L, Jongman RHG, Mucher CA, Bauch B, de Blust G, Parr TW, Olsvig-Whittaker L (2013) The significance of habitats as indicators of biodiversity and their links to

- species. *Ecological Indicators* **33**, 19–25. doi:10.1016/j.ecolind.2012.07.014
- Bürkner P-C (2017) brms: An R Package for Bayesian Multilevel Models using Stan. *Journal of Statistical Software* **80**, 1–28. doi:10.18637/jss.v080.i01
- Caughley J (1985) Effect of fire on the reptile fauna of mallee. In 'Biology of Australian Frogs and Reptiles'. (Eds G Grigg, R Shine, H Ehmann) pp. 31–34. (Royal Zoological Society of NSW and Surrey Beatty and Sons: Chipping Norton, NSW, Australia)
- Clarke MF, Kelly LT, Avitabile SC, Benshemesh J, Callister KE, Driscoll DA, Ewin P, Giljohann K, Haslem A, Kenny SA, Leonard S (2021) Fire and its interactions with other drivers shape a distinctive, semi-arid 'Mallee' ecosystem. *Frontiers in Ecology and Evolution* **9**, 647557. doi:10.3389/fevo.2021.647557
- Cogger HG (1969) A study of the ecology and biology of the mallee dragon (*Amphibolurus fordii*) and its adaptations to survival in an arid environment. PhD Thesis, Macquarie University, Sydney, Australia.
- Cogger HG (1974) Thermal relations of the mallee dragon *Amphibolurus fordii* (Lacertilia: Agamidae). *Australian Journal of Zoology* **22**, 319–339. doi:10.1071/ZO9740319
- Connell J, Watson SJ, Taylor RS, Avitabile SC, Schedvin N, Schneider K, Clarke MF (2019) Future fire scenarios: predicting the effect of fire management strategies on the trajectory of high-quality habitat for threatened species. *Biological Conservation* **232**, 131–141. doi:10.1016/j.biocon.2019.02.004
- Connell J, Hall MA, Nimmo DG, Watson SJ, Clarke MF (2022) Fire, drought and flooding rains: the effect of climatic extremes on bird species' responses to time since fire. *Diversity and Distributions* **28**(3), 417–438. doi:10.1111/ddi.13287
- Craig MD, Withers PC, Bradshaw SD (2006) Patterns of diet and microhabitat use by four species of sympatric *Ctenotus* lizards: do they reveal foraging specialization?. *Journal of the Royal Society of Western Australia* **89**, 1–5.
- Davis RA, Craig MD (2024) Long-term post-fire succession of reptiles in an urban remnant in south-western Australia. *International Journal of Wildland Fire* **33**(6), WF24033. doi:10.1071/WF24033
- Dixon KM, Cary GJ, Worboys GL, Gibbons P (2018) The disproportionate importance of long-unburned forests and woodlands for reptiles. *Ecology and Evolution* **8**(22), 10952–10963. doi:10.1002/ece3.4561
- Doherty TS, Johnson B, Friend GR, Wayne AF (2024a) Multi-year responses of reptiles to prescribed burning in a eucalypt forest ecosystem. *Austral Ecology* **49**(8), e13572. doi:10.1111/aec.13572
- Doherty TS, Bohórquez Fandiño DF, Watchorn DJ, Legge SM, Dickman CR (2024b) Experimentally testing animal responses to prescribed fire size and severity. *Conservation Biology* **38**(3), e14231. doi:10.1111/cobi.14231
- Driscoll DA, Henderson MK (2008) How many common reptile species are fire specialists? A replicated natural experiment highlights the predictive weakness of a fire succession model. *Biological Conservation* **141**, 460–471. doi:10.1016/j.biocon.2007.10.016
- Driscoll DA, Lindenmayer DB, Bennett AF, Bode M, Bradstock RA, Cary GJ, Clarke MF, Dexter N, Fensham R, Friend G, Gill M (2010) Fire management for biodiversity conservation: key research questions and our capacity to answer them. *Biological Conservation* **143**, 1928–1939. doi:10.1016/j.biocon.2010.05.026
- Driscoll DA, Smith AL, Blight S, Maindonald J (2012) Reptile responses to fire and the risk of post-disturbance sampling bias. *Biodiversity and Conservation* **21**, 1607–1625. doi:10.1007/s10531-012-0267-5
- Driscoll DA, Armenteras D, Bennett AF, Brotons L, Clarke MF, Doherty TS, et al. (2021) How fire interacts with habitat loss and fragmentation. *Biological Reviews* **96**, 976–998. doi:10.1111/brv.12687
- Edwards GP, Allan GE, Brock C, Duguid A, Gabrys K, Vaarzon-Morel P (2008) Fire and its management in central Australia. *The Rangeland Journal* **30**, 109–121. doi:10.1071/RJ07037
- Farquhar JE, Russell W, Chapple DG (2023) Identifying the abiotic factors that determine the inland range limits of a mesic-adapted lizard species. *Integrative and Comparative Biology* **64**, 55–66. doi:10.1093/icb/icad124
- Fox BJ (1982) Fire and mammalian secondary succession in an Australian coastal heath. *Ecology* **63**, 1332–1341. doi:10.2307/1938861
- Fox BJ, Taylor JE, Thompson PT (2003) Experimental manipulation of habitat structure: a retrogression of the small mammal succession. *Journal of Animal Ecology* **72**, 927–940. doi:10.1046/j.1365-2656.2003.00765.x
- French BJ, Murphy BP, Bowman DM (2024) Promoting optimal habitat availability by maintaining fine-grained burn mosaics: a modelling study in an Australian semi-arid temperate woodland. *Fire* **7**(6), 172. doi:10.3390/fire7060172
- Friend GR (1993) Impact of fire on small vertebrates in mallee woodlands and heathlands of temperate Australia: a review. *Biological Conservation* **65**, 99–114. doi:10.1016/0006-3207(93)90439-8
- Fyfe G (1980) The effect of fire on lizard communities in central Australia. *Herpetofauna* **12**, 1–9.
- Garden JG, McAlpine CA, Possingham HP, Jones DN (2007) Habitat structure is more important than vegetation composition for local-level management of native terrestrial reptile and small mammal species living in urban remnants: a case study from Brisbane, Australia. *Austral Ecology* **32**, 669–685. doi:10.1111/j.1442-9993.2007.01750.x
- González TM, González-Trujillo JD, Muñoz A, Armenteras D (2022) Effects of fire history on animal communities: a systematic review. *Ecological Processes* **11**(1), 11. doi:10.1186/s13717-021-00357-7
- Greenberg CH, Neary DG, Harris LD (1994) Effect of high-intensity wildfire and silvicultural treatments on reptile communities in sand-pine scrub. *Conservation Biology* **8**, 1047–1057. doi:10.1046/j.1523-1739.1994.08041047.x
- Greene RSB, Chartres CJ, Hodgkinson KC (1990) The effects of fire on the soil in a degraded semiarid woodland. I. Cryptogam cover and physical and micromorphological properties. *Soil Research* **28**(5), 755–777. doi:10.1071/SR9900755
- Greenville AC, Dickman CR (2009) Factors affecting habitat selection in a specialist fossorial skink. *Biological Journal of the Linnean Society* **97**, 531–544. doi:10.1111/j.1095-8312.2009.01241.x
- Griffiths AD, Garnett ST, Brook BW (2015) Fire frequency matters more than fire size: testing the pyrodiversity-biodiversity paradigm for at-risk small mammals in an Australian tropical savanna. *Biological Conservation* **186**, 337–346. doi:10.1016/j.biocon.2015.03.021
- Haslem A, Kelly LT, Nimmo DG, Watson SJ, Kenny SA, Taylor RS, Avitabile SC, Callister KE, Spence-Bailey LM, Clarke MF, Bennett AF (2011) Habitat or fuel? Implications of long-term, post-fire dynamics for the development of key resources for fauna and fire. *Journal of Applied Ecology* **48**, 247–256. doi:10.1111/j.1365-2664.2010.01906.x
- He T, Lamont BB, Pausas JG (2019) Fire as a key driver of Earth's biodiversity. *Biological Reviews* **94**, 1983–2010. doi:10.1111/brv.12544
- Henle K (1990) Population ecology and life history of three terrestrial geckos in arid Australia. *Copeia* **1990**, 759. doi:10.2307/1446442
- Hu Y, Urlus J, Gillespie G, Letnic M, Jessop TS (2013) Evaluating the role of fire disturbance in structuring small reptile communities in temperate forests. *Biodiversity and Conservation* **22**, 1949–1963. doi:10.1007/s10531-013-0519-z
- Keith DA (2012) Functional traits: their roles in understanding and predicting biotic responses to fire regimes from individuals to landscapes. In 'Flammable Australia: Fire Regimes, Biodiversity and Ecosystems in a Changing World'. (Eds RA Bradstock, MA Gill, JE Williams) pp. 97–125. (CSIRO Publishing: Melbourne, Vic, Australia)
- Kelly LT, Nimmo DG, Spence-Bailey LM, Clarke MF, Bennett AF (2010) The short-term responses of small mammals to wildfire in semiarid mallee shrubland, Australia. *Wildlife Research* **37**(4), 293–300. doi:10.1071/WR10016
- Kelly LT, Nimmo DG, Spence-Bailey LM, Haslem A, Watson SJ, Clarke MF, Bennett AF (2011) Influence of fire history on small mammal distributions: insights from a 100-year post-fire chronosequence. *Diversity and Distributions* **17**(3), 462–473. doi:10.1111/j.1472-4642.2011.00754.x
- Lazzari J, Sato CF, Driscoll DA (2022) Traits influence reptile responses to fire in a fragmented agricultural landscape. *Landscape Ecology* **37**(9), 2363–2382. doi:10.1007/s10980-022-01417-9
- Lenth R (2025) emmeans: Estimated Marginal Means, aka Least-Squares Means. R package version 1.11.1-00001. Available at <https://rvinth.github.io/emmeans/>
- Letnic M, Dickman CR, Tischler MK, Tamayo B, Beh CL (2004) The responses of small mammals and lizards to post-fire succession and rainfall in arid Australia. *Journal of Arid Environments* **59**, 85–114. doi:10.1016/j.jaridenv.2004.01.014
- Lindenmayer DB, Wood JT, MacGregor C, Michael DR, Cunningham RB, Crane M, Montague-Drake R, Brown D, Muntz R, Driscoll DA



- (2008) How predictable are reptile responses to wildfire? *Oikos* **117**, 1086–1097. doi:10.1111/j.0030-1299.2008.16683.x
- Loyn RH, Kennedy SJ (2009) Designing old forest for the future: old trees as habitat for birds in forests of Mountain Ash *Eucalyptus regnans*. *Forest Ecology and Management* **258**, 504–515. doi:10.1016/j.foreco.2009.01.005
- Masters P (1996) The effects of fire-driven succession on reptiles in spinifex grasslands at Uluru National Park, Northern Territory. *Wildlife Research* **23**, 39–48. doi:10.1071/WR9960039
- Miritis V, Dickman CR, Nimmo DG, Doherty TS (2024) After the 'Black Summer' fires: faunal responses to megafire depend on fire severity, proportional area burnt and vegetation type. *Journal of Applied Ecology* **61**(1), 63–75. doi:10.1111/1365-2664.14545
- Monamy V, Fox BJ (2000) Small mammal succession is determined by vegetation density rather than time elapsed since disturbance. *Austral Ecology* **25**, 580–587. doi:10.1111/j.1442-9993.2000.tb00063.x
- Morton SR, James CD (1988) The diversity and abundance of lizards in arid Australia: a new hypothesis. *American Naturalist* **132**, 237–256. doi:10.1086/284847
- Moyo S (2022) Community responses to fire: a global meta-analysis unravels the contrasting responses of fauna to fire. *Earth* **3**, 1087–1111. doi:10.3390/earth3040063
- Nimmo DG, Kelly LT, Spence-Bailey LM, Watson SJ, Haslem A, White JG, Clarke MF, Bennett AF (2012) Predicting the century-long post-fire responses of reptiles. *Global Ecology and Biogeography* **21**, 1062–1073. doi:10.1111/j.1466-8238.2011.00747.x
- Nimmo DG, Kelly LT, Spence-Bailey LM, Watson SJ, Taylor RS, Clarke MF, Bennett AF (2013) Fire mosaics and reptile conservation in a fire-prone region. *Conservation Biology* **27**, 345–353. doi:10.1111/j.1523-1739.2012.01958.x
- Nimmo DG, Kelly LT, Farnsworth LM, Watson SJ, Bennett AF (2014) Why do some species have geographically varying responses to fire history? *Ecography* **37**, 805–813. doi:10.1111/ecog.00684
- Nimmo DG, Avitabile S, Banks SC, Bliege Bird R, Callister K, Clarke MF, Dickman CR, Doherty TS, Driscoll DA, Greenville AC, Haslem A (2019) Animal movements in fire-prone landscapes. *Biological Reviews* **94**(3), 981–998. doi:10.1111/brv.12486
- Nimmo DG, Andersen AN, Archibald S, Boer MM, Brotons L, Parr CL, Tingley MW (2022) Fire ecology for the 21st century. *Diversity and Distributions* **28**(3), 350–356. doi:10.1111/ddi.13482
- Oksanen J (2024) Vegan: an introduction to ordination. Available at <https://cran.r-project.org/web/packages/vegan/vignettes/intro-vegan.pdf> [verified 1 December 2024].
- Olson AR, Farquhar JE, Bettio MK (2018) *Ctenotus regius* (Regal Striped Skink). Diet. *Herpetological Review* **49**(2), 328–329.
- Partridge DA, Lewis T, Tran CT, Castley JG (2023) Fire and habitat variables explain reptile community abundance and richness in subtropical open eucalypt forests. *International Journal of Wildland Fire* **32**, 1089–1108. doi:10.1071/WF22001
- Pausas JG, Keeley JE (2009) A burning story: the role of fire in the history of life. *BioScience* **59**, 593–601. doi:10.1525/bio.2009.59.7.10
- Pianka ER (1969) Sympatry of desert lizards (*Ctenotus*) in Western Australia. *Ecology* **50**, 1012–1030. doi:10.2307/1936893
- Pianka ER, Goodyear SE (2012) Lizard responses to wildfire in arid interior Australia: Long-term experimental data and commonalities with other studies. *Austral Ecology* **37**, 1–11. doi:10.1111/j.1442-9993.2010.02234.x
- R Development Core Team (2024) R: A language and environment for statistical computing. Version 3.3.3. (R Foundation for Statistical Computing: Vienna) Available at <https://www.r-project.org/> [verified 1 December 2024].
- Rice B, Westoby M (1999) Regeneration after fire in *Triodia* R. Br. *Australian Journal of Ecology* **24**, 563–572. doi:10.1046/j.1442-9993.1999.01004.x
- Rodda G (1978) 1975 bushfires in Northern Scotia country and their aftermath. *Range Management Newsletter* **78**, 4–5.
- Sadler RA, Colgan D, Beatson CA, Cogger HG (2019) *Ctenophorus spinodomus* sp. nov., a new species of dragon lizard (Squamata: Agamidae) from *Triodia* mallee habitat of Southeast Australia. *Records of the Australian Museum* **71**, 199–215. doi:10.3853/j.2201-4349.71.2019.1700
- Sass S, Wilson A (2006) Effects of fire on lizard communities in the mallee shrublands of western New South Wales. *Herpetofauna* **36**, 106–108.
- Schlesinger CA, Noble JC, Weir T (1997) Fire studies in mallee (*Eucalyptus* spp.) communities of western New South Wales: reptile and beetle populations in sites of differing fire history. *The Rangeland Journal* **19**(2), 190–205. doi:10.1071/RJ9970190
- Smith AL (2018) Successional changes in trophic interactions support a mechanistic model of post-fire population dynamics. *Oecologia* **186**(1), 129–139. doi:10.1007/s00442-017-4016-z
- Smith AL, Bull CM, Driscoll DA (2013) Successional specialization in a reptile community cautions against widespread planned burning and complete fire suppression. *Journal of Applied Ecology* **50**, 1178–1186. doi:10.1111/1365-2664.12119
- Sousa WP (1984) The role of disturbance in natural communities. *Annual Review of Ecology and Systematics* **15**, 353–391. doi:10.1146/annurev.es.15.110184.002033
- Tews J, Brose U, Grimm V, Tielbörger K, Wichmann MC, Schwager M, Jeltsch F (2004) Animal species diversity driven by habitat heterogeneity/diversity: the importance of keystone structures. *Journal of Biogeography* **31**, 79–92. doi:10.1046/j.0305-0270.2003.00994.x
- Valentine LE, Reaveley A, Johnson B, Fisher R, Wilson BA (2012) Burning in banksia woodlands: how does the fire-free period influence reptile communities? *PLoS One* **7**, e34448. doi:10.1371/journal.pone.0034448
- Verdon SJ, Watson SJ, Nimmo DG, Clarke MF (2020) Are all fauna associated with the same structural features of the foundation species *Triodia scariosa*? *Austral Ecology* **45**(6), 773–787. doi:10.1111/aec.12894
- von Takach B, Jolly CJ, Dixon KM, Penton CE, Doherty TS, Banks SC (2022) Long-unburnt habitat is critical for the conservation of threatened vertebrates across Australia. *Landscape Ecology* **37**(6), 1469–1482. doi:10.1007/s10980-022-01427-7
- Watson SJ, Taylor RS, Nimmo DG, Kelly LT, Haslem A, Clarke MF, Bennett AF (2012) Effects of time since fire on birds: how informative are generalized fire response curves for conservation management? *Ecological Applications* **22**, 685–696. doi:10.1890/11-0850.1
- Westbrooke ME (2010) 'Nanya Station Western New South Wales: Conservation Research Education.' (University of Ballarat: Vic, Australia)
- Westbrooke ME (2012) The pastoral history, biological and cultural significance of the Scotia Country, far western New South Wales. *Proceedings of the Linnean Society of New South Wales* **134**, A55–A68.
- Westbrooke ME, Miller JD, Kerr MKC (1998) Vegetation of the Scotia 1:100 000 Map Sheet, Western New South Wales. *Cunninghamia* **5**, 685–684.
- Withers M (1989) 'The Bushmen of the Great Anabranch.' (Withers: Adelaide, SA, Australia)
- Woinarski JC, Burbidge AA, Harrison PL (2015) Ongoing unraveling of a continental fauna: decline and extinction of Australian mammals since European settlement. *Proceedings of the National Academy of Sciences* **112**(15), 4531–4540. doi:10.1073/pnas.1417301112
- York A, Bell TL, Weston CJ (2012) Fire regimes and soil-based ecological processes: implications for biodiversity. In 'Flammable Australia: Fire Regimes, Biodiversity and Ecosystems in a Changing World'. (Eds RA Bradstock, MA Gill, JE Williams) pp. 127–148. (CSIRO Publishing: Melbourne, Vic, Australia)



**Data availability.** Data and R code used in this study are available on the Monash University Bridges data repository: [10.26180/29532416](https://bridges.monash.edu/10.26180/29532416)

**Conflicts of interest.** The authors have no conflicts of interests to declare. Simon Cook (author deceased): no conflicts of interest available.

**Declaration of funding.** No specific funding was received for this work.

**Acknowledgements.** We thank Federation University for managing Nanya Station and facilitating our research on the property.

**Author affiliations**

<sup>A</sup>School of Biological Sciences, Monash University, Clayton, Vic 3800, Australia.

<sup>B</sup>Future Regions Research Centre, Federation University Australia, Gippsland, Vic 3353, Australia.

## Appendix 1. Abundance data for all small terrestrial vertebrates trapped in each fire age (years since last fire (YSLF))

| Class    | Family           | Species                                                | 1 YSLF | 18 YSLF | >40 YSLF | Total |
|----------|------------------|--------------------------------------------------------|--------|---------|----------|-------|
| Reptilia | Carphodactylidae | <i>Nephrurus levis</i>                                 | 3      | 5       | 3        | 11    |
| Reptilia | Diplodactylidae  | <i>Diplodactylus vittatus</i>                          | 0      | 0       | 1        | 1     |
| Reptilia | Diplodactylidae  | <i>Lucasium damaeum</i>                                | 15     | 4       | 1        | 20    |
| Reptilia | Diplodactylidae  | <i>Rhynchoedura angusta</i>                            | 11     | 3       | 0        | 14    |
| Reptilia | Diplodactylidae  | <i>Strophurus elderi</i> <sup>VU</sup>                 | 0      | 5       | 1        | 6     |
| Reptilia | Diplodactylidae  | <i>Strophurus williamsi</i>                            | 0      | 2       | 0        | 2     |
| Reptilia | Gekkonidae       | <i>Gehyra versicolor</i>                               | 0      | 0       | 1        | 1     |
| Reptilia | Pygopodidae      | <i>Delma australis</i> <sup>EN</sup>                   | 0      | 1       | 0        | 1     |
| Reptilia | Pygopodidae      | <i>Lialis burtonis</i>                                 | 1      | 1       | 1        | 3     |
| Reptilia | Scincidae        | <i>Ctenotus atlas</i>                                  | 8      | 15      | 11       | 34    |
| Reptilia | Scincidae        | <i>Ctenotus brachyonyx</i>                             | 7      | 0       | 7        | 14    |
| Reptilia | Scincidae        | <i>Ctenotus regius</i>                                 | 30     | 1       | 0        | 31    |
| Reptilia | Scincidae        | <i>Ctenotus schomburgkii</i>                           | 3      | 2       | 7        | 12    |
| Reptilia | Scincidae        | <i>Ctenotus taeniatus</i>                              | 0      | 15      | 1        | 16    |
| Reptilia | Scincidae        | <i>Cyclodomorphus melanops elongatus</i> <sup>EN</sup> | 1      | 0       | 1        | 2     |
| Reptilia | Scincidae        | <i>Lerista aericeps</i>                                | 3      | 3       | 1        | 7     |
| Reptilia | Scincidae        | <i>Lerista labialis</i>                                | 23     | 6       | 0        | 29    |
| Reptilia | Scincidae        | <i>Lerista punctatovittata</i>                         | 1      | 1       | 6        | 8     |
| Reptilia | Scincidae        | <i>Liopholis inornata</i>                              | 0      | 2       | 4        | 6     |
| Reptilia | Scincidae        | <i>Menetia greyii</i>                                  | 2      | 2       | 5        | 9     |
| Reptilia | Scincidae        | <i>Morethia boulengeri</i>                             | 0      | 0       | 1        | 1     |
| Reptilia | Agamidae         | <i>Ctenophorus pictus</i>                              | 3      | 0       | 0        | 3     |
| Reptilia | Agamidae         | <i>Ctenophorus spinodomus</i>                          | 62     | 31      | 20       | 113   |
| Reptilia | Agamidae         | <i>Diporiphora nobbi</i>                               | 6      | 1       | 4        | 11    |
| Reptilia | Agamidae         | <i>Pogona vitticeps</i>                                | 2      | 4       | 3        | 9     |
| Reptilia | Varanidae        | <i>Varanus gouldii</i>                                 | 2      | 2       | 1        | 5     |
| Reptilia | Typhlopidae      | <i>Anilius bicolor</i>                                 | 0      | 2       | 0        | 2     |
| Reptilia | Typhlopidae      | <i>Anilius bituberculatus</i>                          | 1      | 0       | 6        | 7     |
| Reptilia | Elapidae         | <i>Brachyuropsis australis</i>                         | 0      | 1       | 1        | 2     |
| Reptilia | Elapidae         | <i>Demansia psammophis</i>                             | 1      | 0       | 0        | 1     |
| Reptilia | Elapidae         | <i>Suta nigriceps</i>                                  | 0      | 2       | 0        | 2     |
| Reptilia | Elapidae         | <i>Pseudonaja mengdeni</i>                             | 1      | 0       | 1        | 2     |

(Continued on next page)

| Class           | Family          | Species                                    | 1 YSLF | 18 YSLF | >40 YSLF | Total |
|-----------------|-----------------|--------------------------------------------|--------|---------|----------|-------|
| Reptilia        | Elapidae        | <i>Pseudonaja modesta</i> <sup>EN</sup>    | 4      | 2       | 0        | 6     |
| Reptilia        | Elapidae        | <i>Vermicella annulata</i>                 | 0      | 2       | 0        | 2     |
| Amphibia        | Limnodynastidae | <i>Neobatrachus sudellae</i>               | 1      | 0       | 0        | 1     |
| Mammalia        | Dasyuridae      | <i>Ningauai yvonneae</i> <sup>VU</sup>     | 2      | 63      | 64       | 129   |
| Mammalia        | Burramyidae     | <i>Cercartetus concinnus</i> <sup>EN</sup> | 0      | 0       | 1        | 1     |
| Mammalia        | Muridae         | <i>Pseudomys bolami</i> <sup>EN</sup>      | 2      | 1       | 0        | 3     |
| Mammalia        | Muridae         | <i>Mus musculus</i>                        | 1      | 3       | 0        | 4     |
| Total abundance |                 |                                            | 196    | 182     | 153      | 531   |

Each YSLF treatment had four replicate sites. Nomenclatures of reptiles follows the [ASH \(2023\)](#) Official List of Australian Species. The conservation status is presented as 'VU' = Vulnerable and 'EN' = Endangered next to names of species listed as threatened in the state of NSW.